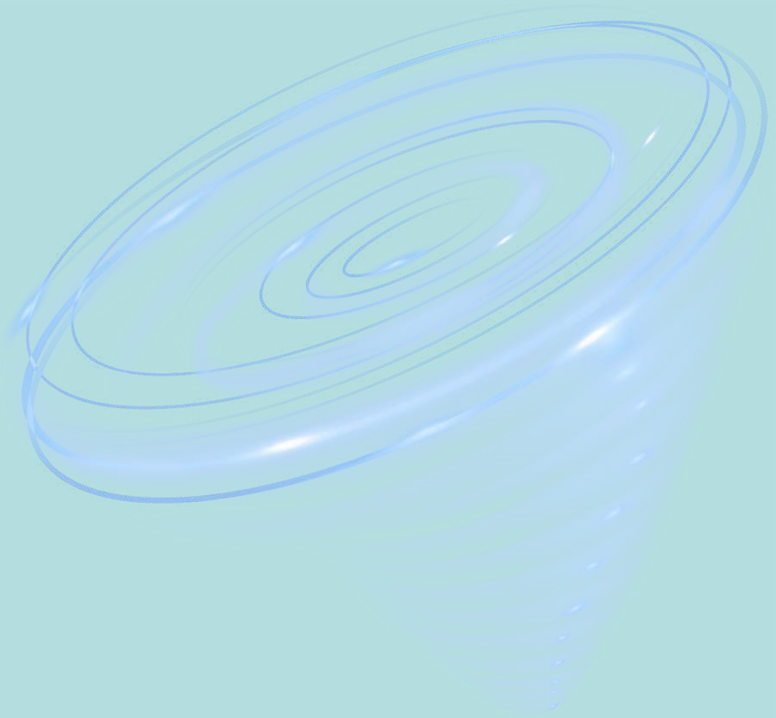
THEME : Les vortex

COMMENT ÉTUDIER PUIS CARACTÉRISER LE MOUVEMENT D'UN VORTEX?

DROIN Emeline

SOMMAIRE

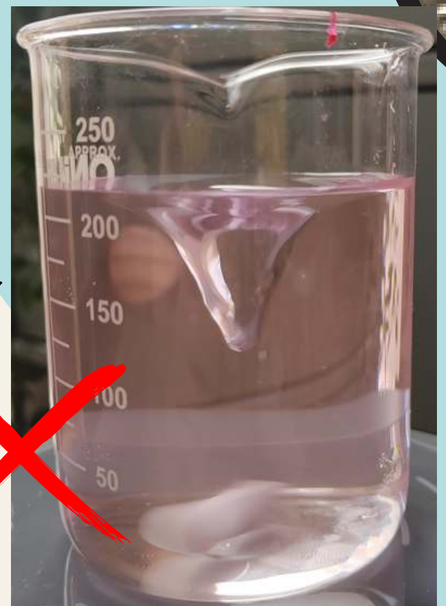
- Création puis observation d’un vortex
- Etude de la trajectoire d’un élément de fluide
- Etude cinématique du vortex
- Conclusion

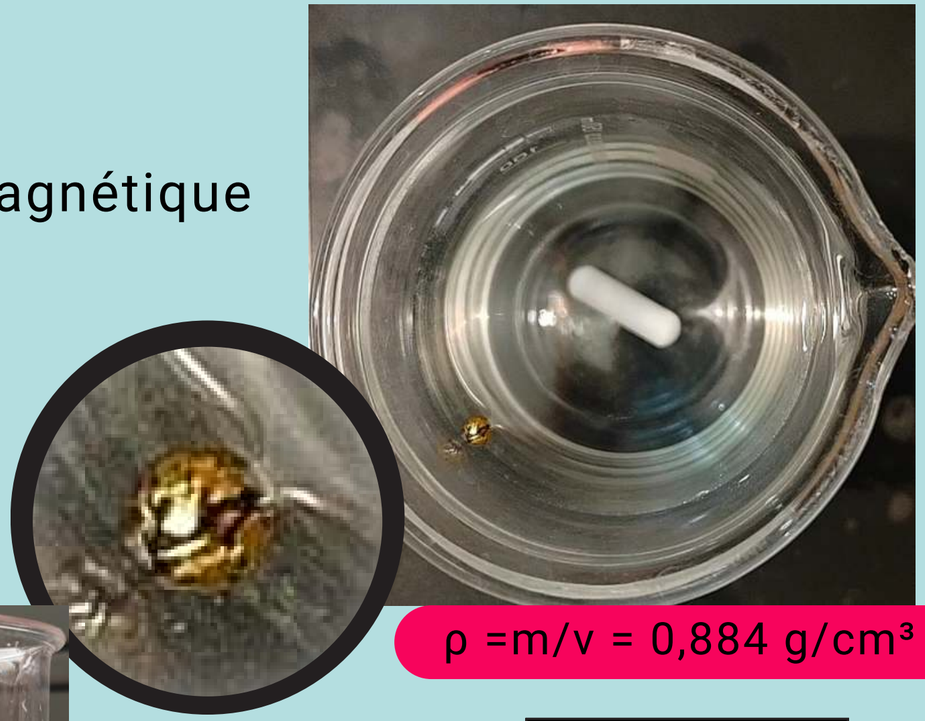


CRÉATION PUIS OBSERVATION D'UN VORTEX :

1-SUIVI DU MOUVEMENT

- observation grâce à l'agitateur magnétique
- suivre un élément de fluide





$\rho = m/v = 0{,}884\ g/cm^3$

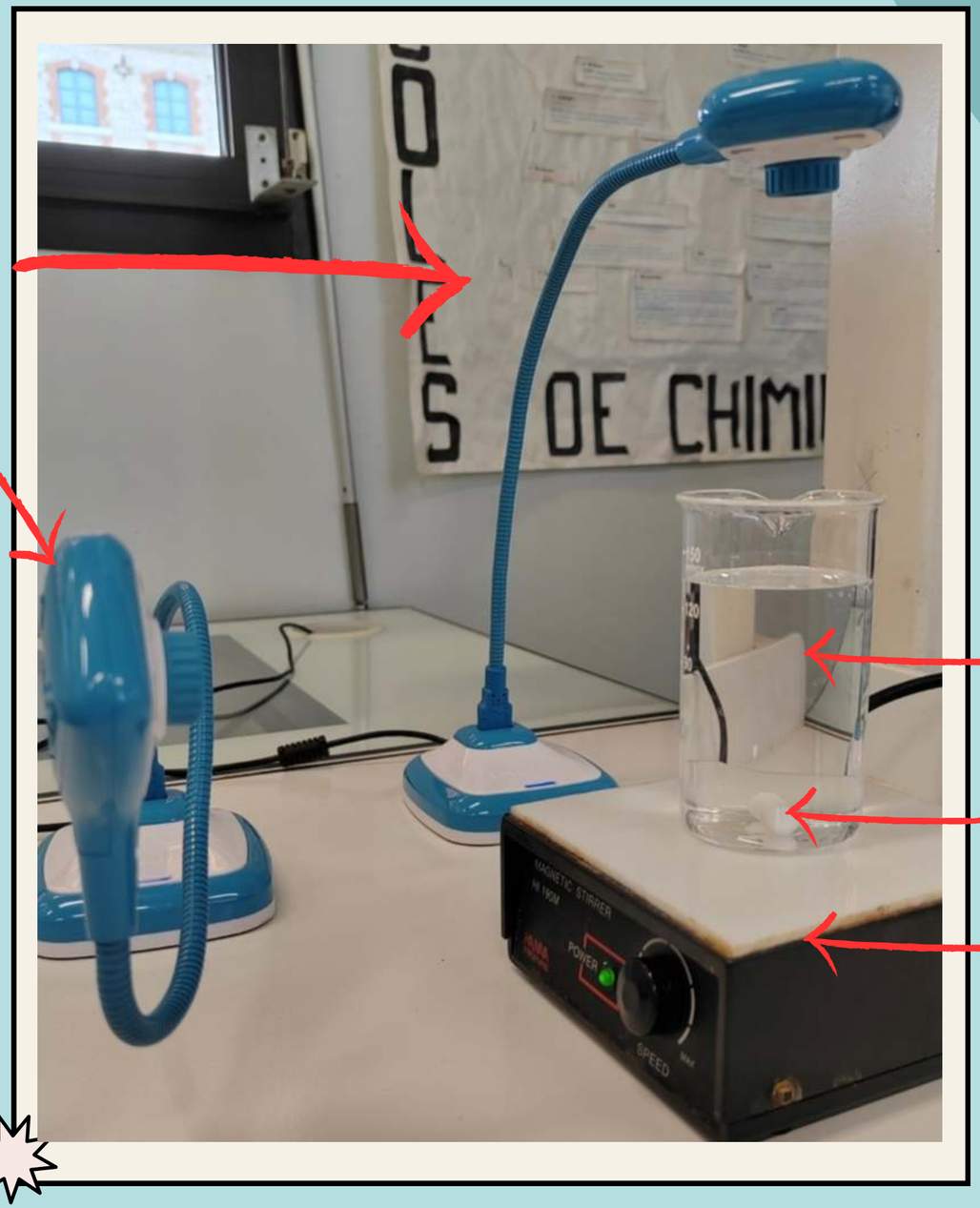
$$u(\rho) = \sqrt{\left(\frac{u(m)}{m}\right)^2 + \left(\frac{u(v)}{v}\right)^2} * \rho$$

$\rho = (0.884 \pm 0.130)\ g/cm^3$

$< \rho_{eau} = 1\ g/cm^3$

2-EXPÉRIENCE RÉALISÉE

caméras
× 2

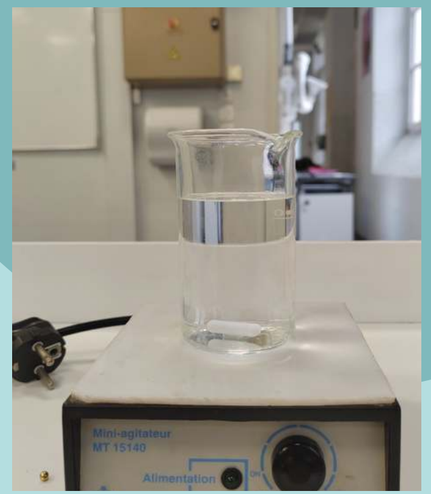


bécher de 150mL
- eau

barreau aimanté

agitateur magnétique

De profil :



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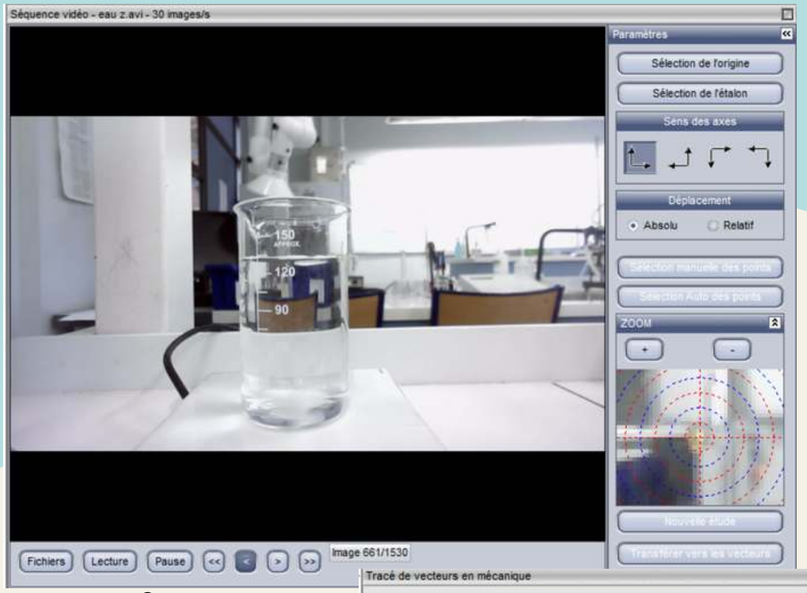
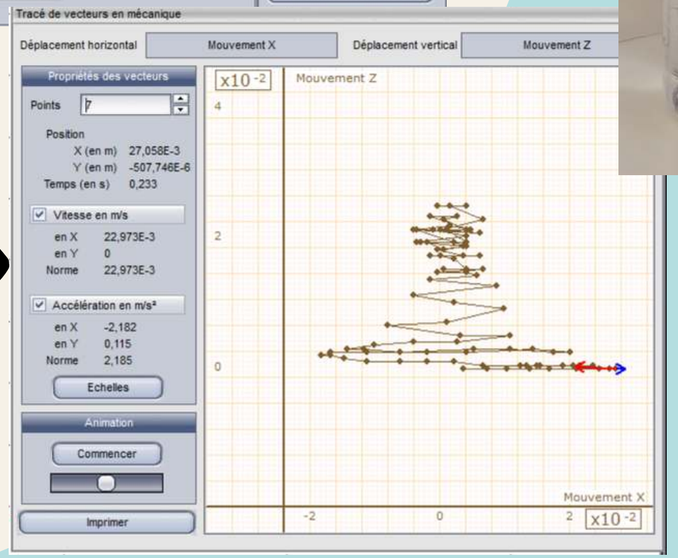
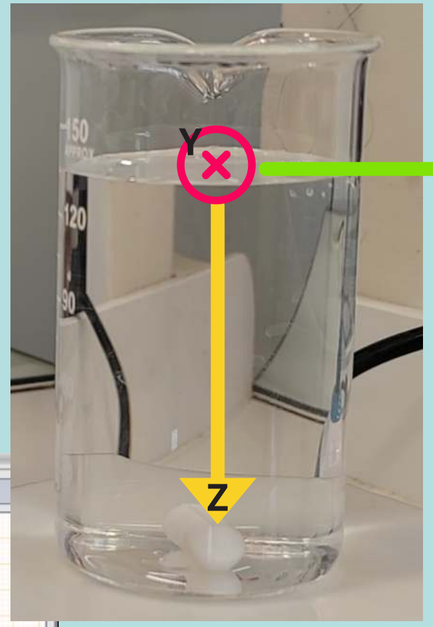
Du dessus :

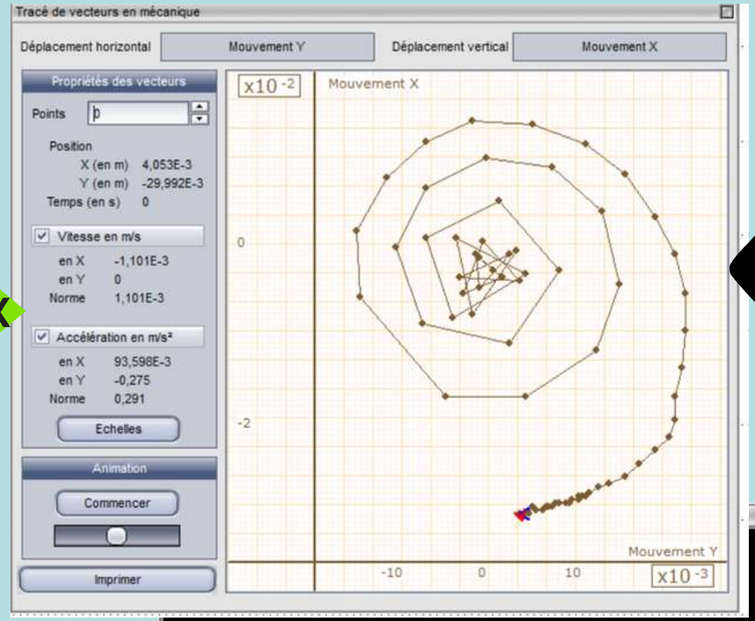
ETUDE DE LA TRAJECTOIRE D'UN ÉLÉMENT DE FLUIDE

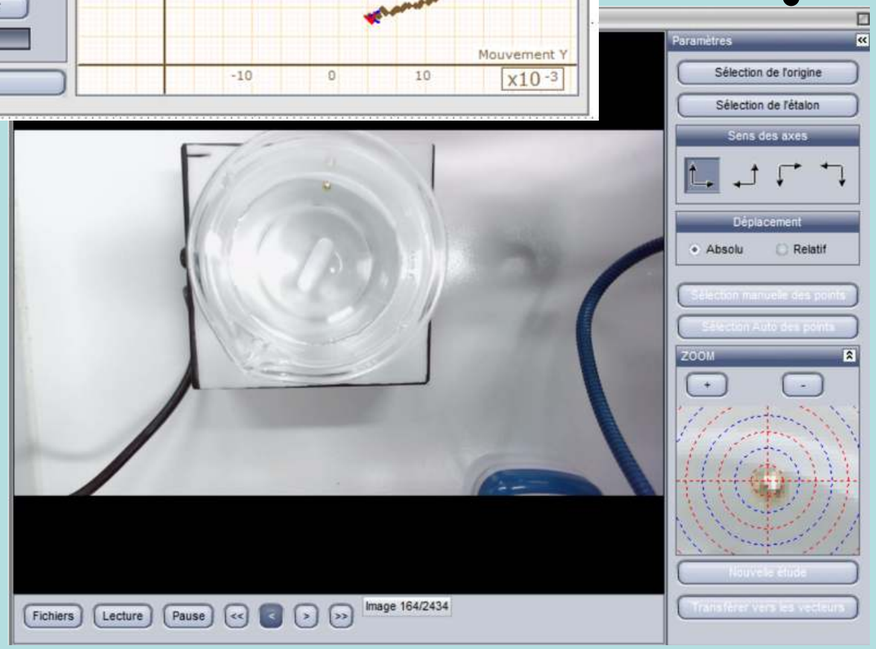
1-POINTAGE VIDÉO ET COORDONNÉE CARTÉSIENNE :







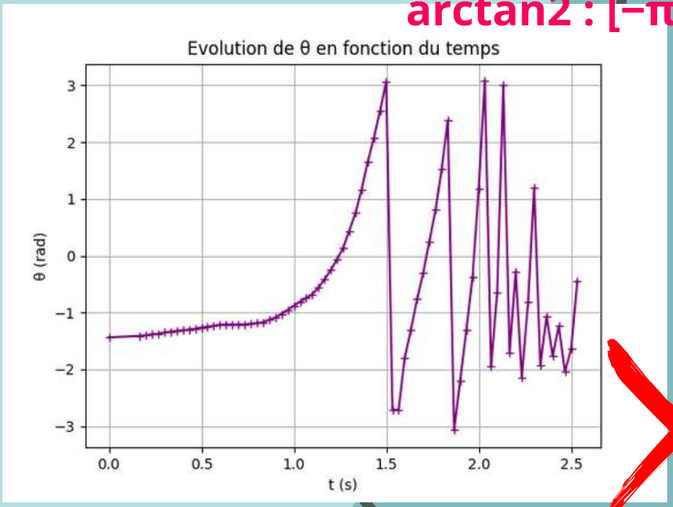


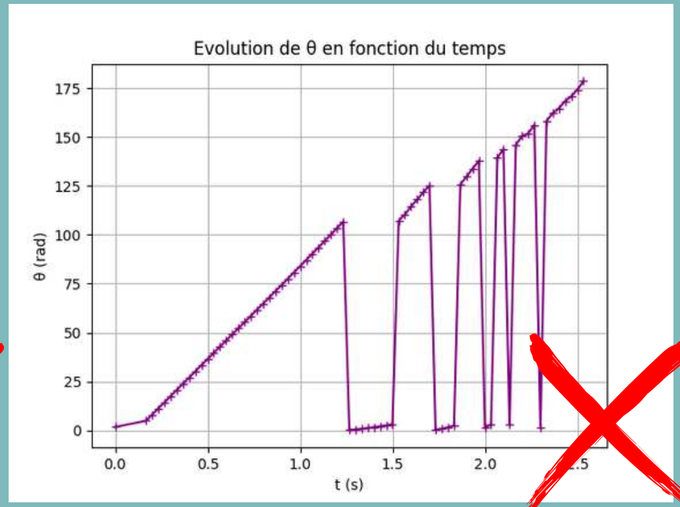


2-CONVERSION COORDONNÉE CYLINDRIQUE :

```
from math import sqrt
from math import atan2
import numpy as np
def convert(a,b):
    Lr=[]
    Ltheta=[]
    for i in range (len(a)):
        r=np.sqrt((a[i]**2)+(b[i]**2))
        Lr.append(r)
        theta=atan2(b[i], a[i])
        Ltheta.append(theta)
    return {"Lr": Lr, "Ltheta": Ltheta}

convert(x,y)
```

arctan2 : [−π,π]

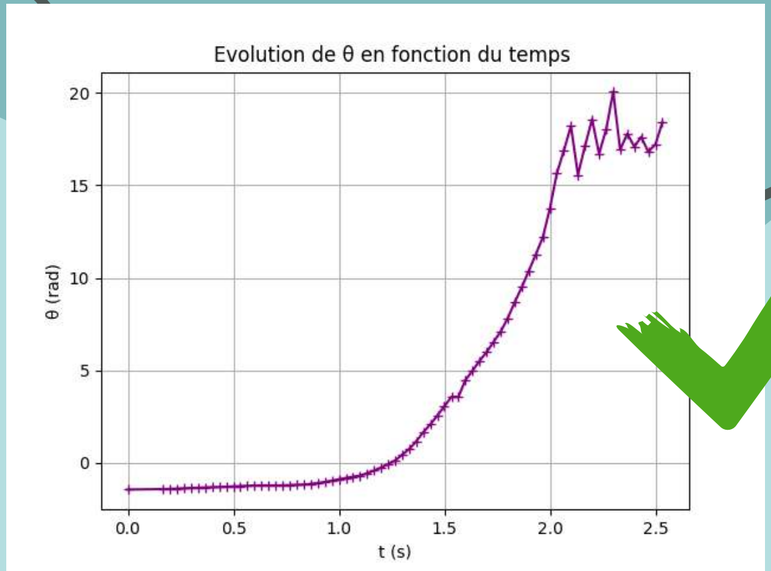





```
theta_corrigé = np.unwrap(theta)
```

Premier essai pour arranger θ :

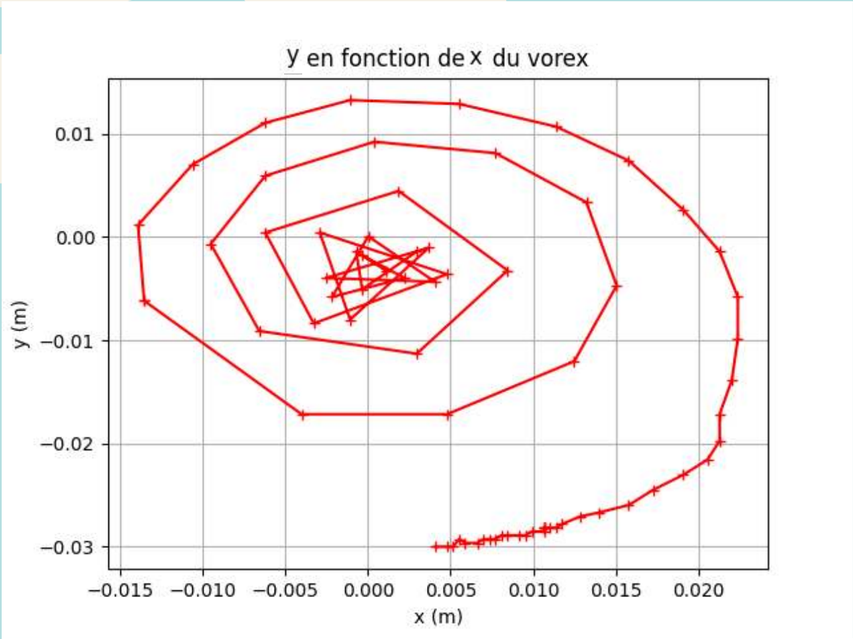
```
def arrangement_theta(L):
    nb_pi=0
    for i in range(len(L)):
        if L[i]<0:
            nb_pi = nb_pi + 1
            L[i] = L[i] + (nb_pi*np.pi)
    return L
arrangement_theta(theta)
```

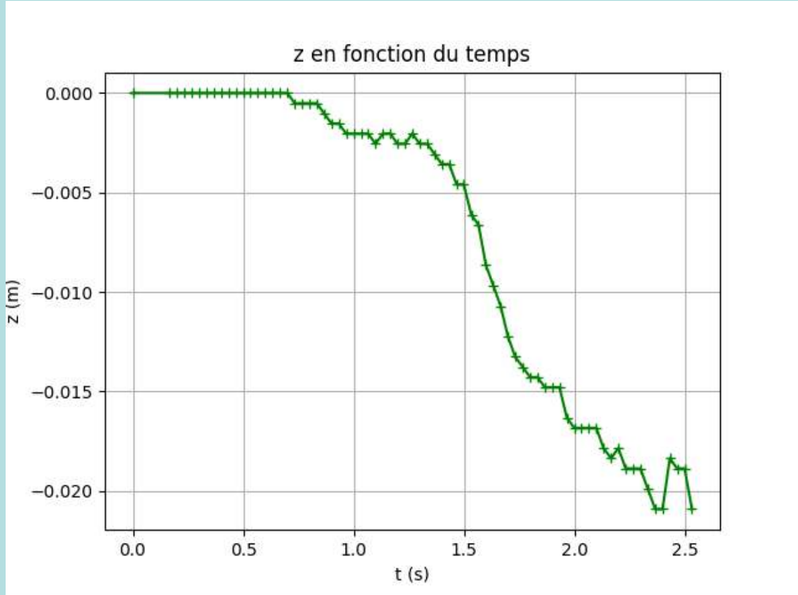


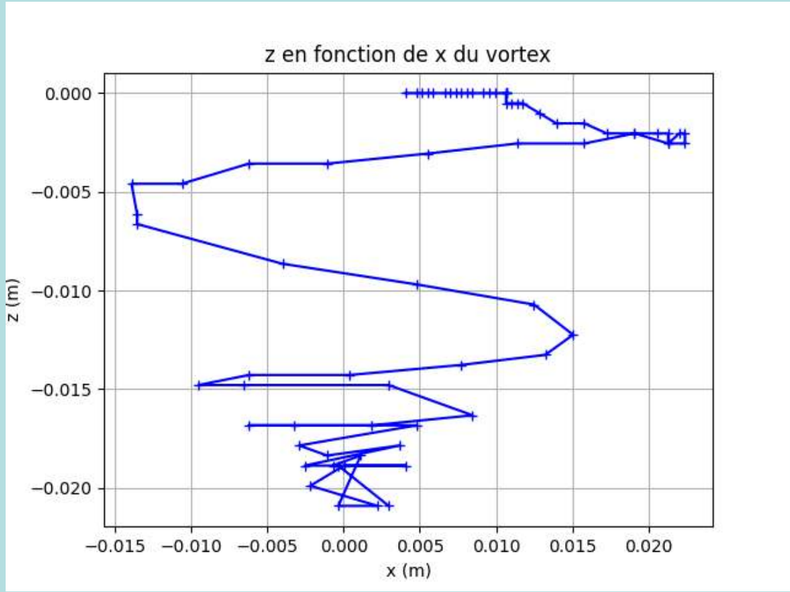
ETUDE CINÉMATIQUE DU VORTEX

1-COORDONNÉE CARTÉSIENNE :

```
# 1)- Courbe x vs y

plt.plot(x,y,'+', color='red')
plt.plot(x,y, color='red')
plt.xlabel('x (m)')
plt.ylabel('y (m)')
plt.title('y en fonction de x du vorex')
plt.grid()
plt.show()

# 2)- Courbe z vs x

plt.plot(x,z,'+', color='blue')
plt.plot(x,z, color='blue')
plt.title('z en fonction de x du vortex')
plt.xlabel('x (m)')
plt.ylabel('z (m)')
plt.grid()
plt.show()

# 3)- Courbe z vs t

plt.plot(t, z, '+', color= 'green')
plt.plot(t, z, color= 'green')
plt.title('z en fonction du temps')
plt.xlabel('t (s)')
plt.ylabel('z (m)')
plt.grid()
plt.show()
```

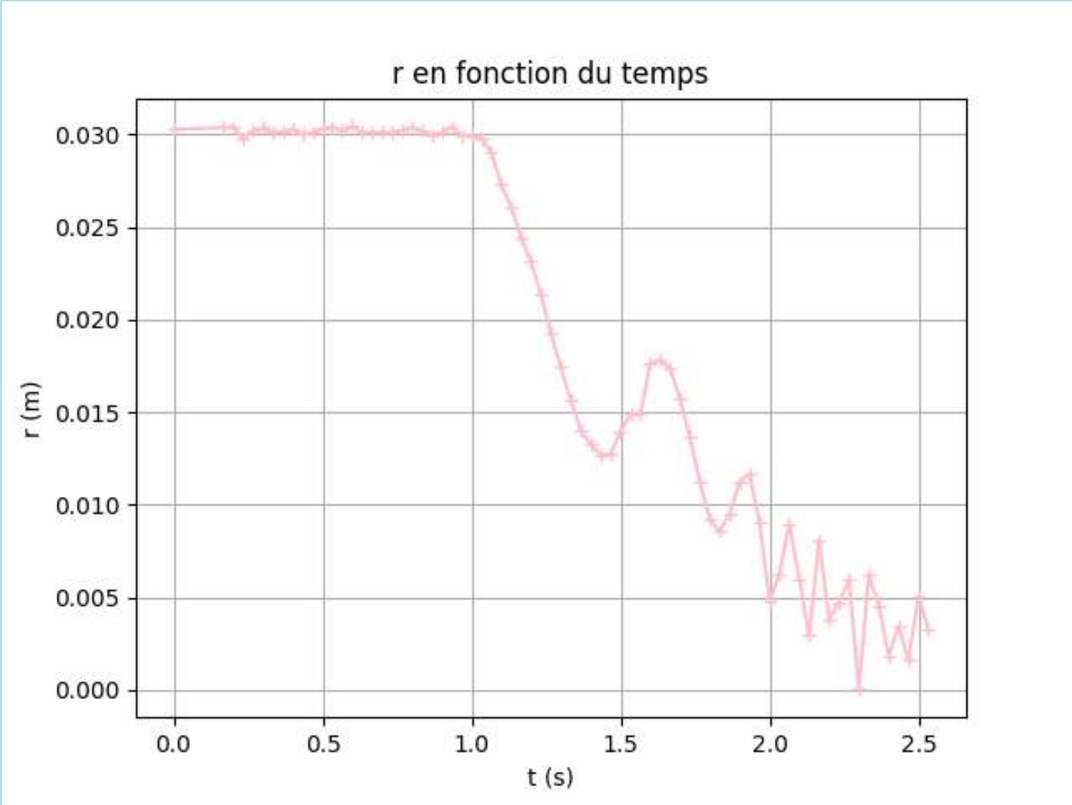




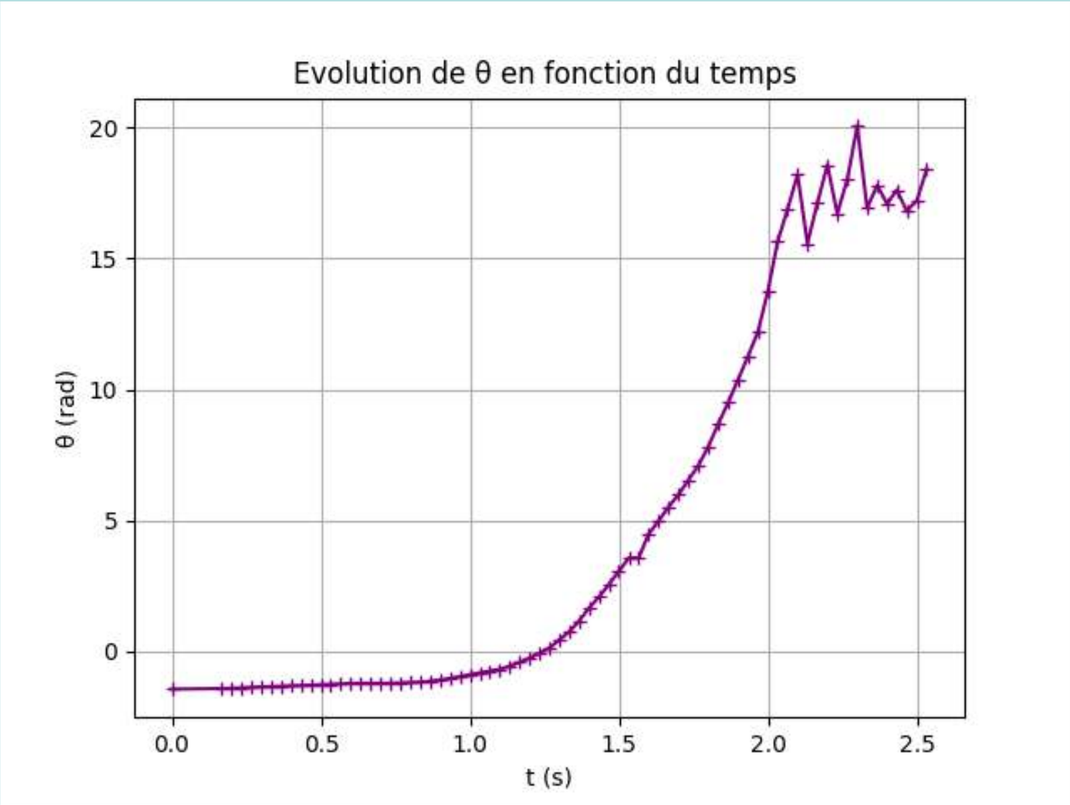


2-COORDONNÉE CYLINDRIQUE:

```
# 4)- Courbe r vs t

plt.plot(t, r, '+', color='pink')
plt.plot(t, r, color='pink')
plt.title('r en fonction du temps')
plt.xlabel('t (s)')
plt.ylabel('r (m)')
plt.grid()
plt.show()
```

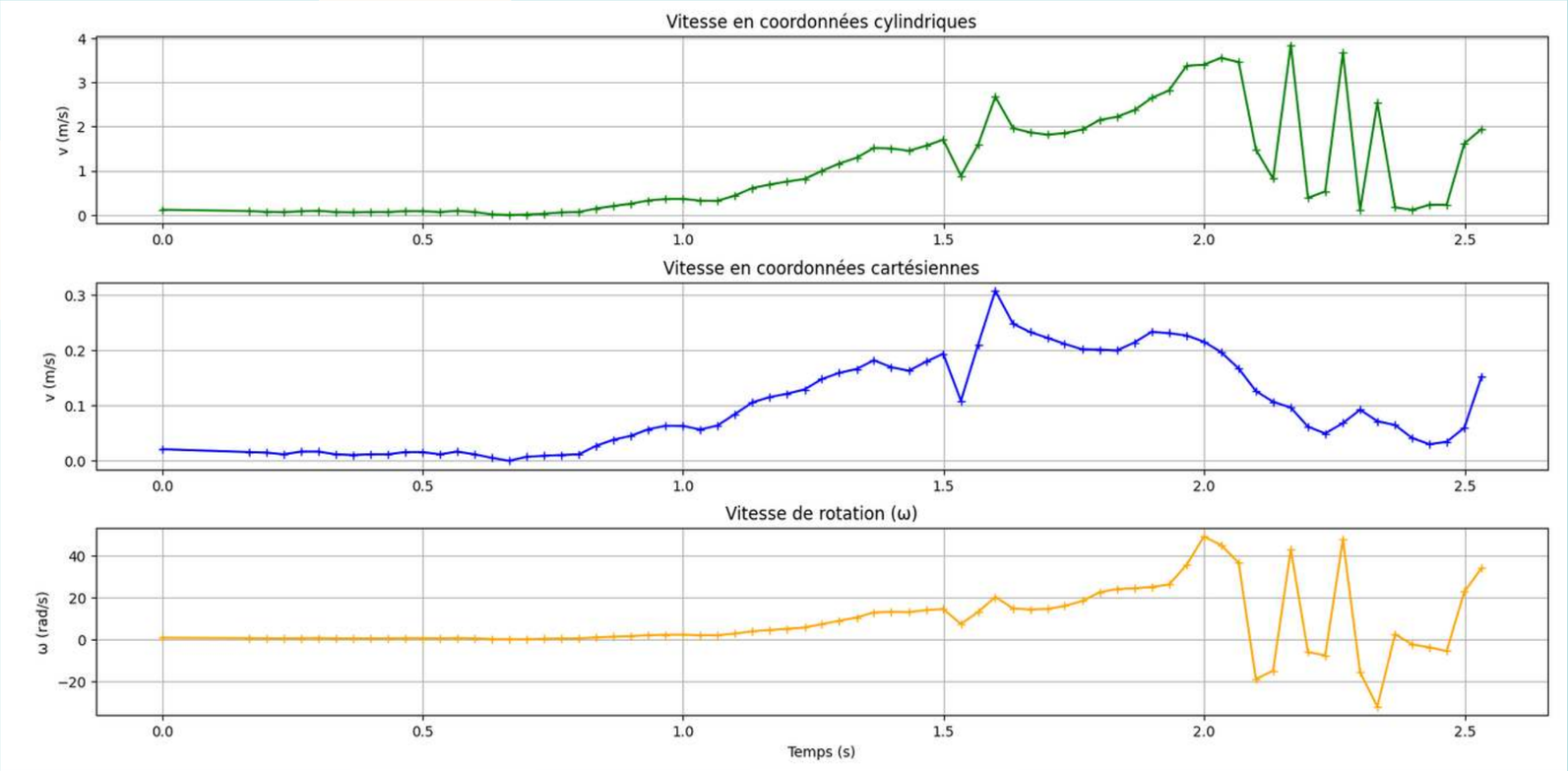



```
## 5)- Courbe theta vs t
plt.plot(t, theta_corrigé, '+', color='purple')
plt.plot(t, theta_corrigé, color='purple')
plt.title('Evolution de θ en fonction du temps ')
plt.xlabel('t (s)')
plt.ylabel('θ (rad)')
plt.grid()
plt.show()
```



3- EVOLUTION DE VITESSE AU COURS DU TEMPS

```python
### III- vitesse mouvement
dt = np.mean(np.diff(t))

# Dérivées
dr_dt = np.gradient(r, dt)
dtheta_dt = np.gradient(theta_corrigé, dt)
vx = np.gradient(x, dt)
vy = np.gradient(y, dt)
vz = np.gradient(z, dt)

# Vitesses
v_cyl = np.sqrt(dr_dt**2 + (r * ((dtheta_dt)**2))+(vz**2))
omega = dtheta_dt
v_cart = np.sqrt(vx**2 + (vy**2)+(vz**2))

## Affichage courbes vitesses
plt.figure(figsize=(12, 8))
# Vitesse cylindrique
plt.subplot(3, 1, 1)
plt.plot(t, v_cyl,'+', color='green')
plt.plot(t, v_cyl, color='green')
plt.ylabel('v (m/s)')
plt.title('Vitesse en coordonnées cylindriques')
plt.grid()

# Vitesse cartésienne
plt.subplot(3, 1, 2)
plt.plot(t, v_cart,'+', color='blue')
plt.plot(t, v_cart, color='blue')
plt.ylabel('v (m/s)')
plt.title('Vitesse en coordonnées cartésiennes')
plt.grid()

# Vitesse de rotation
plt.subplot(3, 1, 3)
plt.plot(t, omega,'+', color='orange')
plt.plot(t, omega, color='orange')
plt.xlabel('Temps (s)')
plt.ylabel('ω (rad/s)')
plt.title('Vitesse de rotation (ω)')
plt.grid()
```



CONCLUSION